Automated Air Pollution Forecasting Using Time Series Models

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***Abstract*—An automated air quality forecasting system for Chennai is developed by integrating time series modeling with real-time data acquisition. Historical pollutant concentrations—$PM_{2.5}$, PM_{10}, CO, NO_2, and SO_2—obtained from the Central Pollution Control Board (CPCB) serve as the foundation for model training. A Seasonal Autoregressive Integrated Moving Average model (SARIMA) is employed, selected after extensive hyperparameter tuning for its superior performance over baseline models. The model is trained on Air Quality Index (AQI) values computed using CPCB's sub-index methodology and is retrained daily to adapt to evolving pollution trends. After optimization, the SARIMA model achieves a Mean Absolute Percentage Error (MAPE) of 13.57%, demonstrating strong predictive accuracy. The system features real-time API integration for continuous data ingestion and automated scheduling for data collection, model retraining, and forecast generation, thereby requiring minimal manual intervention. This scalable and adaptive framework supports short-term air quality forecasting, enabling proactive environmental management and informed public health decision-making.**

***Index Terms*—air quality index, forecasting, time series analysis, SARIMA, automation, environmental monitoring**

I. Introduction

Air pollution remains a pressing environmental and public health concern in many urban areas, particularly in rapidly expanding cities like Chennai, India. Prolonged exposure to elevated levels of pollutants such as $PM_{2.5}$, PM_{10}, CO, NO_2, and SO_2 has been well-documented to cause adverse health effects [4]. Timely and accurate forecasting of air quality is essential for issuing health advisories, implementing pollution control strategies, and supporting data-driven policymaking.

The Air Quality Index (AQI) provides a standardized measure to quantify air pollution levels and communicate potential health risks to the public. Forecasting AQI involves modeling the temporal patterns of pollutant concentrations, which are influenced by complex factors including traffic emissions, meteorological conditions, industrial activity, and seasonal variations. Machine learning techniques, particularly time series models, offer promising approaches to capturing these dynamics and generating reliable short-term predictions.

TABLE I
Pollutant Abbreviations and Their Full Forms

Abbreviation	Full Form
$PM_{2.5}$	Particulate Matter (diameter $\leq$ 2.5µm)
PM_{10}	Particulate Matter (diameter $\leq$ 10µm)
SO_2	Sulfur Dioxide
NO_2	Nitrogen Dioxide
CO	Carbon Monoxide

However, traditional forecasting approaches often struggle to adapt to rapidly changing environmental conditions and typically require manual updates, making them inefficient for continuous operation. Therefore, there is a need for an automated and adaptive system capable of handling real-time data streams, dynamically retraining itself, and maintaining high forecasting accuracy over time.

In this work, we develop an automated AQI forecasting system based on the Seasonal Autoregressive Integrated Moving Average (SARIMA) model. After comparing multiple time series models, SARIMA was selected based on its initial superior performance. Following model selection, extensive hyperparameter tuning was conducted to further optimize its forecasting accuracy. The final SARIMA model achieved a Mean Absolute Percentage Error (MAPE) of 13.57%. Our system integrates real-time pollutant data ingestion via APIs, daily model retraining, and automated forecasting pipelines, ensuring robust and adaptive AQI predictions. Through this framework, we aim to deliver accurate, scalable, and real-time

AQI forecasts to support proactive environmental management and public health decision-making.

II. Literature Review

Accurate forecasting of air quality is critical for safeguarding public health and guiding environmental policies. As air pollution patterns grow increasingly complex, research has transitioned from traditional statistical methods to advanced deep learning and hybrid models capable of modeling temporal dependencies and nonlinear behaviors more effectively.

Rani Samal et al. [5] explored air quality forecasting in Bhubaneswar using SARIMA and Prophet models on data from 2005–2015. They found that Prophet, particularly with log-transformed data, outperformed linear models in terms of RMSE and MSE. Their work demonstrated the potential of time series models in environmental monitoring. Similarly, Bhatti et al. [9] applied SARIMA and factor analysis to $PM_{2.5}$ trends in Lahore, revealing deteriorating air quality and underscoring the need for predictive systems that support timely decision-making.

Deep learning approaches have gained prominence for their ability to handle non-stationary and nonlinear data. Sani et al. [6] implemented a stacked LSTM model for air quality prediction in Dhaka, achieving over 90% accuracy in both short- and long-term forecasting. In a related study, Sofiah et al. [13] compared LSTM and SARIMA models for Gurugram, concluding that LSTM delivered superior results, particularly in capturing nonlinear AQI fluctuations. Yang et al. [12] introduced a Seasonal Gated Recurrent Unit (SGRU) enhanced with Kalman filtering for pollution forecasting in Taiwan, effectively addressing seasonal effects and noise in the data. This approach has influenced our own motivation to adopt models that adapt to AQI volatility.

The fusion of classical and neural models has also shown promising results. Yang et al. [7] proposed a hybrid SARIMA–LSTM model for energy forecasting, improving prediction accuracy by modeling residuals from statistical outputs. Likewise, Chen et al. [8] built an LSTM–SARIMA hybrid system to forecast energy demands, incorporating contextual data like weather and server load. While our system does not use exogenous variables, these studies validate the benefits of hybrid architectures for capturing both linear trends and nonlinear dynamics. Manikandan et al. [10] paired SARIMA with Gaussian Process Regression (GPR) to model uncertainty in urban air pollution forecasting, highlighting the need for models that generalize across diverse urban contexts.

Recent developments have also focused on spatial dynamics and contextual awareness. Terroso-Saenz et al. [14] used Heterogeneous Graph Neural Networks (GNNs) to incorporate spatial and sensor data for more accurate forecasting. Kempaiah et al. [11] examined AQI variations across India, showing strong dependencies on time-of-day and meteorological conditions.

Collectively, the literature reflects a shift from purely statistical forecasting to more adaptive, hybrid, and data-driven approaches. Inspired by these advancements, our system integrates Prophet, ARIMA, and SARIMA models in a unified, retrainable pipeline. It pulls real-time data via APIs and automates daily updates, enabling a scalable and responsive forecasting solution tailored for urban environments—without relying on exogenous variables.

III. System Architecture

The proposed system (Fig. 1) is designed to deliver an automated, scalable, and highly accurate air quality forecasting solution tailored for urban environments. The architecture integrates both classical time series models and modern machine learning techniques in a unified pipeline that operates in two major phases: training and deployment.

In the training phase, historical air quality data from credible sources are preprocessed, which includes handling missing values and aggregating pollutants. Feature engineering steps, including lag feature creation and seasonality decomposition, are applied to capture the temporal dynamics of pollutant levels.

Multiple forecasting models—such as ARIMA, SARIMA, SARIMA, and Prophet—are trained on the processed data. Each model is evaluated using metrics like RMSE and MAE to select the best-performing configurations. Hyperparameter tuning is performed to optimize model accuracy, and hybrid strategies are adopted where residuals from classical models are corrected using machine learning regressors.

In the deployment phase, the best-performing model ensemble is integrated with a live API that continuously fetches the latest pollution and meteorological data. The system is configured to retrain models periodically to adapt to evolving pollution patterns and maintain forecasting accuracy over time. Forecast outputs are postprocessed to generate daily AQI values, along with pollutant-specific trends, and can be visualized through an integrated dashboard interface.

IV. Methodology

The dataset used in this study was sourced from the Central Pollution Control Board (CPCB) [3], spanning from January 2021 to January 2025, and includes hourly concentrations for pollutants such as $PM_{2.5}$, PM_{10}, CO, NO_2, SO_2, and O_3. The dataset contains 27 features, including pollutant concentrations, meteorological data, and temporal indicators in Chennai. Preprocessing steps were performed to prepare the data for time-series forecasting. Invalid entries were removed, and pollutant concentrations were standardized to a common unit (e.g., converting CO from mg/m^3 to μg/m^3). Additionally, missing values were handled through forward-fill imputation to maintain continuity in the time series. After preprocessing, the data was retained in its hourly time-series format for further analysis.

To calculate the Air Quality Index (AQI), sub-index statistics were computed for each pollutant in accordance with the CPCB guidelines. The sub-index for a given pollutant p is determined using the following equation:

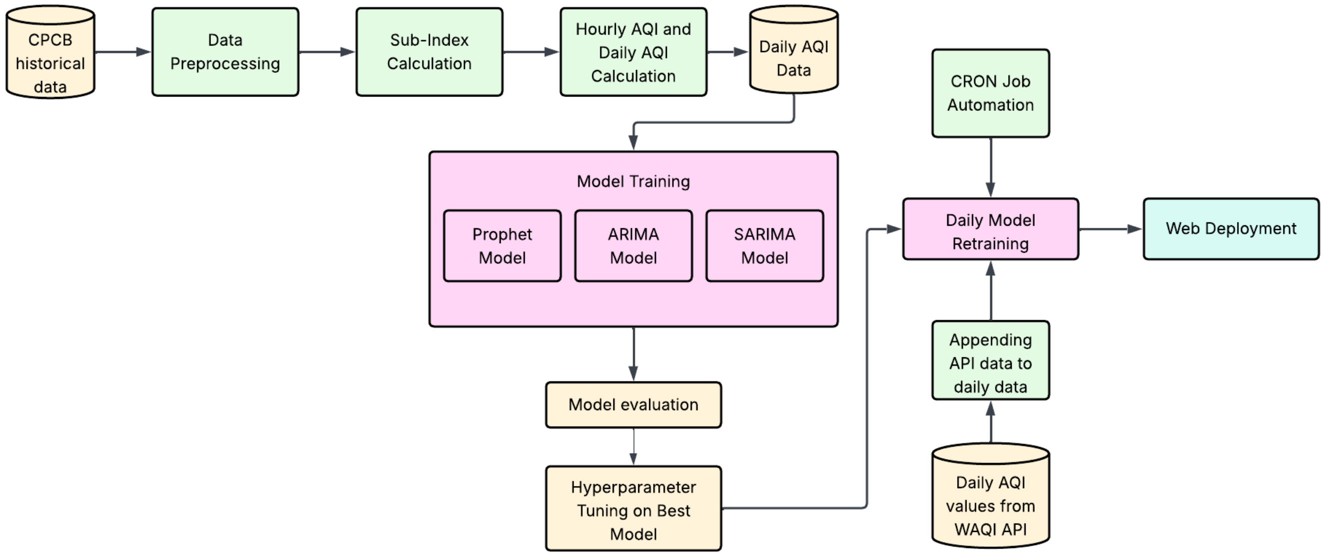

Fig. 1. System Architecture for AQI Forecasting

$$I_p = \left(\frac{I_{Hi} - I_{Lo}}{BP_{Hi} - BP_{Lo}} \right) \times (C_p - BP_{Lo}) + I_{Lo}$$

where I_p is the sub-index for pollutant p, C_p is the concentration of pollutant p, BP_{Hi} and BP_{Lo} are the upper and lower breakpoints for the pollutant, and I_{Hi} and I_{Lo} are the AQI breakpoints corresponding to these concentrations. The overall AQI for each time point is then derived by selecting the maximum sub-index from all pollutants. This approach ensures that the AQI reflects the pollutant with the highest concentration, which is most critical for air quality at that specific time.

For time-series forecasting, three models were considered: Prophet, ARIMA, and SARIMA.

A. PROPHET

The Prophet model, developed by Facebook, is designed to handle time-series data with strong seasonal patterns. The model decomposes the time series into trend, seasonality, and holiday components, as represented by the following equation:

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t$$

where $y(t)$ is the observed value at time t, $g(t)$ is the trend component, $s(t)$ is the seasonal component, $h(t)$ represents holiday effects, and ϵ_t is the error term. Prophet is particularly effective for handling data with strong seasonal and holiday effects, but may not capture all complex relationships in pollutant concentrations.

Prophet uses piecewise linear or logistic growth models for the trend component, and a Fourier series to model seasonal effects, enabling it to flexibly capture multiple seasonalities in the data.

B. ARIMA

The ARIMA model, or Autoregressive Integrated Moving Average, predicts future values based on past observations. The model is given by:

$$\begin{aligned} Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \phi_p Y_{t-p} \\ + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \cdots + \theta_q \epsilon_{t-q} + \epsilon_t \end{aligned}$$

where Y_t is the observed value at time t, $\phi_1, \ldots, \phi_p$ are the autoregressive parameters, $\theta_1, \ldots, \theta_q$ are the moving average parameters, and ϵ_t is the error term. The ARIMA model was initially configured with an order of (2, 1, 2) for the non-seasonal parameters. The choice of these parameters was informed by exploratory data analysis, where we observed significant autocorrelation in the time series data. Specifically, an autoregressive order of 2 ($p = 2$) and a moving average order of 2 ($q = 2$) were selected to effectively capture the underlying temporal dependencies, balancing model complexity and performance. The first-order differencing ($d = 1$) was applied to achieve stationarity in the series, which is crucial for the ARIMA model's predictive accuracy. While ARIMA was capable of capturing the linear trends and dependencies, it was noted that the model was less adept at accounting for seasonal fluctuations inherent in the AQI data.

C. SARIMA

Recognizing the seasonal patterns in the AQI data, the SARIMA model was configured with an order of (1, 1, 1) for the non-seasonal components and (1, 1, 1, 7) for the seasonal components. The seasonal period was set to 7 ($s = 7$) to capture weekly patterns, which were identified through visual analysis and autocorrelation plots, as air quality tends to exhibit weekly cycles due to factors like traffic and industrial activity. The SARIMA model's inclusion of both seasonal and non-seasonal components allows it to better handle the multi-faceted dependencies in the AQI data. The selected parameters (1, 1, 1) for both the seasonal and non-seasonal parts are relatively simple but effective, providing a good balance between capturing essential patterns and avoiding overfitting. The SARIMA model outperformed ARIMA in handling the seasonal fluctuations and external variables, which made it more suitable for AQI forecasting.

$$Y_t = \mu + \phi_1 Y_{t-1} + \cdots + \phi_p Y_{t-p} + \theta_1 \epsilon_{t-1} + \cdots + \theta_q \epsilon_{t-q} + \sum_{i=1}^{P} \Phi_i Y_{t-s} + \cdots + \sum_{j=1}^{Q} \Theta_j \epsilon_{t-s} + \epsilon_t$$

where Y_t is the observed value at time t, μ is the intercept, $\phi_1, \ldots, \phi_p$ are the autoregressive parameters, $\theta_1, \ldots, \theta_q$ are the moving average parameters, $\Phi_1, \ldots, \Phi_P$ are the seasonal autoregressive parameters, $\Theta_1, \ldots, \Theta_Q$ are the seasonal moving average parameters, and ϵ_t is the error term. The inclusion of both seasonal and non-seasonal components allows SARIMA to effectively model both short-term fluctuations and long-term dependencies in the AQI data.

For model evaluation, the performance of each model was assessed using standard metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE). These metrics were computed for both training and test datasets to evaluate model accuracy and generalizability. SARIMA, with its ability to model both seasonality and trends, was selected as the best-performing model for AQI forecasting.

D. Hyperparameter Tuning

Hyperparameter optimization for the SARIMA model was performed using grid search. The grid search considered the following range of hyperparameters for the non-seasonal and seasonal components:

$$(p, d, q) \in \{0, 1, 2, ...15\}, \quad (P, D, Q) \in \{0, 1, 2, 3\}, \quad s = 7$$

The hyperparameter tuning ensures that the model can effectively handle the non-stationary and seasonal nature of the AQI data.

To ensure the model remains up to date, the SARIMA model is retrained daily after receiving new AQI data via an API. The latest AQI observations are appended to the existing dataset, and the model is retrained at three specific times each day: 5 AM, 1 PM, and 10 PM. This approach allows the model to continuously incorporate the most recent air quality data and improve forecasting accuracy over time.

V. Results and Discussion

The models were trained using data spanning from January 2, 2021, to December 31, 2024, which consisted of 1460 daily observations. The testing set was drawn from January 1, 2025, to March 31, 2025, covering 90 observations. The primary goal was to predict AQI values for the testing period based on the historical data. To evaluate model performance, three key metrics were used: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE).

The ARIMA(2,1,2) model, when applied to the AQI data (Fig. 2), provided reasonable forecasting performance. The MAE value of 19.1693 indicates that, on average, the model's predictions deviated from the actual AQI values by approximately 19 units. The RMSE of 22.0456 suggests that, while the model performed adequately, larger prediction errors occurred more frequently. The MAPE of 29.8500% further confirms that the ARIMA model struggles with maintaining accuracy across different levels of AQI values, particularly when dealing with high levels of variability in the data.

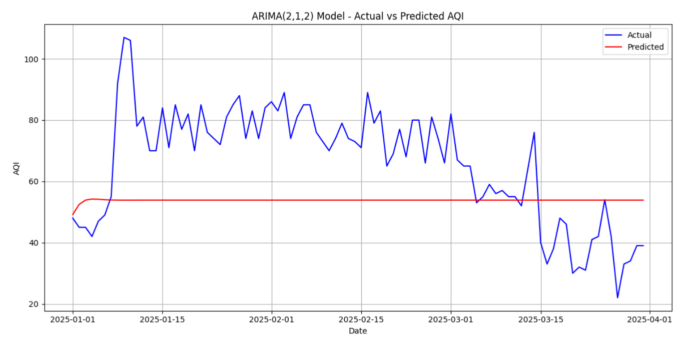

Fig. 2. ARIMA Model: Actual vs Predicted AQI Values

The SARIMA(1,1,1)(1,1,1,7) model (Fig. 3) achieved improved performance over the ARIMA model. The MAE of 18.0194 indicates that the SARIMA model's predictions were, on average, closer to the actual AQI values. The RMSE of 20.7107 further illustrates that the model successfully reduced the magnitude of larger prediction errors compared to ARIMA. The MAPE of 28.9064% highlights the model's better accuracy, especially for medium-to-low AQI values. The seasonal component of SARIMA, combined with the non-seasonal ARIMA structure, allowed the model to capture both short-term fluctuations and long-term trends in AQI data.

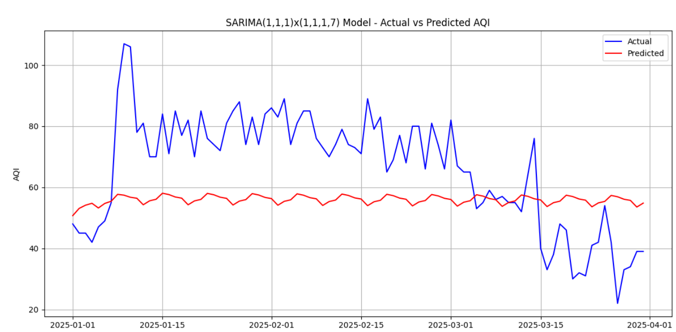

Fig. 3. SARIMA Model: Actual vs Predicted AQI Values

The Prophet model (Fig. 4), designed to handle seasonality and trends, showed a higher degree of error. With an MAE of

26.5433, the Prophet model's predictions were less accurate compared to the ARIMA and SARIMA models. The RMSE of 29.6840 suggests that the model struggled with larger prediction errors, while the MAPE of 39.0652% demonstrates that the Prophet model was less effective in predicting AQI values accurately across the testing period. Although Prophet incorporates flexible trend adjustments and seasonality modeling, the errors indicate that it might not be as well-suited for forecasting AQI data in the context of this study.

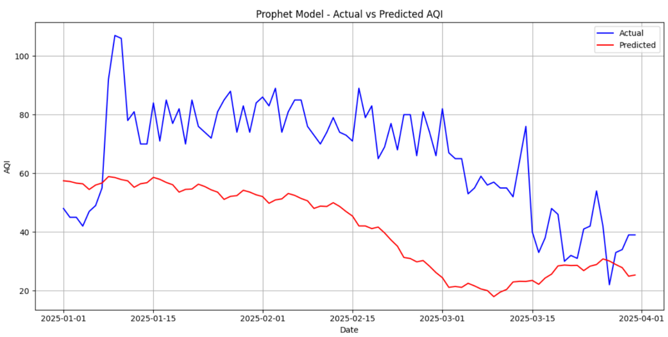

Fig. 4. Prophet Model: Actual vs Predicted AQI Values

Despite the performance differences, the results of the SARIMA model, after tuning, demonstrate its superiority for this specific time-series forecasting task. The ability of SARIMA to capture both seasonal and non-seasonal components in the data proved to be beneficial in generating accurate predictions. This model's success underlines the importance of selecting the correct forecasting model for time-series data with complex seasonal patterns, like AQI, which can vary based on time of day, week, and even season.

VI. Model Comparison

TABLE II
Model Performance Comparison

Metric	ARIMA	SARIMA	Prophet
MAE	19.1693	18.0194	26.5433
RMSE	22.0456	20.7107	29.6840
MAPE	29.8500%	28.9064%	39.0652%

The performance of the ARIMA, SARIMA, and Prophet models was evaluated using three key metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE). These metrics provide a comprehensive assessment of the models' ability to predict AQI values accurately. Table II summarizes the results for all three models.

From the results (Fig. 5), it is evident that the SARIMA model outperformed both the ARIMA and Prophet models across all three metrics. Specifically, the SARIMA model achieved the lowest MAE of 18.0194, followed by ARIMA with an MAE of 19.1693. The Prophet model exhibited the highest MAE, with a value of 26.5433, indicating larger prediction errors on average.

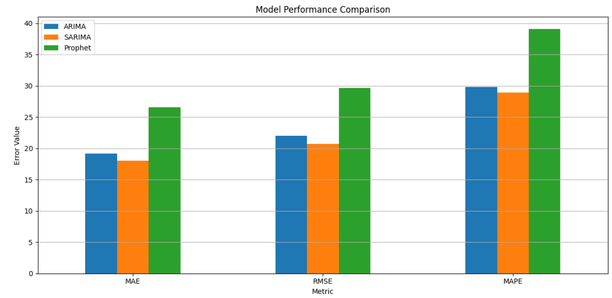

Fig. 5. Model Performance Comparison with Metrics

The RMSE values further reinforce the comparison, with SARIMA again showing the lowest value of 20.7107, suggesting fewer large errors in its predictions. ARIMA followed with an RMSE of 22.0456, while Prophet exhibited the highest RMSE at 29.6840, highlighting its struggle with large prediction errors.

Similarly, the MAPE values indicate that SARIMA provided the most accurate predictions relative to the actual AQI values, with a MAPE of 28.9064%. ARIMA's MAPE of 29.8500% showed slightly higher errors, while Prophet's MAPE of 39.0652% suggests a considerable discrepancy between the predicted and actual values.

The performance comparison suggests that SARIMA, with its ability to capture both seasonal and non-seasonal components, provides the most reliable forecasts for AQI data. The flexibility of SARIMA in handling complex patterns in time-series data makes it the most suitable model for this forecasting task. Although Prophet's flexibility with trend changes is beneficial in some contexts, its performance on AQI data indicates that it might not be the best fit for this particular application. Similarly, while ARIMA performed reasonably well, its inability to incorporate seasonal trends as effectively as SARIMA limits its forecasting accuracy. Hyperparameter tuning was performed on the SARIMA model to improve accuracy, and the optimal configuration identified was $(p, d, q) = (0, 3, 15)$ and $(P, D, Q, s) = (0, 0, 0, 7)$, chosen based on cross-validation performance.

Thus, based on the evaluation metrics, SARIMA emerges as the best model for AQI forecasting in this study.

A. Hyperparameter Tuning of SARIMA

After selecting SARIMA as the most promising model based on initial comparison, hyperparameter tuning was conducted to further enhance its predictive performance. The tuning process involved varying the non-seasonal and seasonal parameters (p, d, q) and (P, D, Q, s) to minimize forecasting errors. The evolution of error metrics during the tuning process is illustrated in Fig. 6. The final tuned configuration was determined to be $(p, d, q) = (0, 3, 15)$ and $(P, D, Q, s) = (0, 0, 0, 7)$. After tuning, the SARIMA model achieved the following improved metrics on the training set, as summarized in Table II.

TABLE III
SARIMA Model Metrics After Hyperparameter Tuning

Metric	Value
MAE	11.7992
RMSE	19.4125
MAPE	13.5732%

These results demonstrate a significant improvement compared to the untuned model, confirming the effectiveness of hyperparameter optimization.

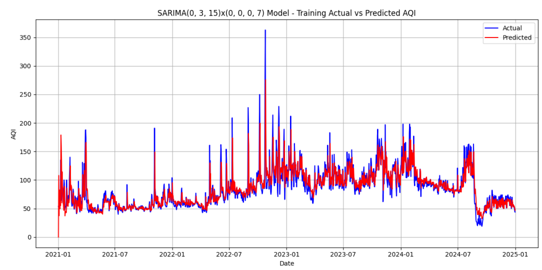

Fig. 6. SARIMA after Hyperparameter Tuning

VII. Conclusion

This study presents a robust air quality forecasting system that integrates real-time pollutant monitoring, CPCB-compliant AQI computation, and advanced time series forecasting. Multiple models, including Prophet, ARIMA, and SARIMA, were evaluated, with SARIMA emerging as the best performer due to its ability to capture seasonality and leverage external variables. After hyperparameter tuning, SARIMA achieved a MAPE of 13.57%, ensuring high forecasting accuracy. Daily retraining further enhanced adaptability to changing environmental conditions. The developed framework offers a scalable and efficient solution for short-term AQI forecasting, suitable for deployment across diverse urban environments.

VIII. Future Scope

Future work will focus on incorporating meteorological variables to further improve predictive performance and exploring deep learning models to capture more complex temporal patterns. These enhancements aim to strengthen the system's capability for proactive air quality management.

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